

HITHA SHAMASUNDAR

Tempe, AZ (Willing to Relocate) | <http://www.linkedin.com/in/hitha-s> | hshamasu@asu.edu | [hitha-shamasundar-portfolio](#) | +1 602-785-0597

Master of Science in Computer Software Engineering graduate with industry experience developing AI-powered applications, retrieval-augmented generation (RAG) systems, scalable backend services, and full-stack web applications. Seeking Software Engineer & AI Engineer opportunities to build intelligent and scalable software solutions while contributing across the design, development, testing, and deployment lifecycle.

EDUCATION

Master of Science in Software Engineering

Arizona State University (GPA: 3.93/4.0)

May 2026

Tempe, AZ

Bachelor of Engineering in Computer Science and Engineering

NIE Institute of Technology - Visvesvaraya Technological University (CGPA: 8.18/10.0)

August 2023

Bengaluru, India

SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, C++

Backend, Microservices & Web: Flask, Node.js, NestJS, Next.js, React, REST APIs, GraphQL, Vue, HTML, CSS

Database: MySQL, PostgreSQL, NoSQL

DevOps, Cloud & Testing: AWS, Docker, Kubernetes, Git, Jenkins, CI/CD, Selenium, Cypress, Cucumber, Postman, REST Assured, TestNG, JUnit

AI & Machine Learning: LangChain, OpenAI API, Gemini API, FAISS, ChromaDB, TensorFlow, OpenCV, YOLO

EXPERIENCE

HR Service Delivery Operations - Student Worker, Office of Human Resources – ASU, Tempe AZ

August 2025 - Present

- Designed and implemented workflow automation solutions using **Google Apps Script (JavaScript)** and **Google Sheets APIs**, streamlining scheduling and compliance-tracking processes, reducing manual effort by approximately 70%, and improving operational efficiency.
- Supported a large-scale digitization initiative involving 25,000+ employee records by developing staffing models, throughput forecasts, dashboards, and project tracking tools that enabled data-driven planning, resource allocation, and project execution.
- Conducted pilot testing, analyzed operational data, and collaborated with cross-functional stakeholders to identify process bottlenecks, optimize workflows, and deliver scalable solutions that improved reporting visibility and business operations.

Software Engineer (SDE/SDET), Happiest Minds Technologies Limited, Bengaluru, IN

August 2023 - June 2024

- Architected and developed DocBot, an AI-powered conversational application using **Retrieval-Augmented Generation (RAG)**, **LangChain**, **OpenAI** embeddings, FAISS, and semantic retrieval techniques including document parsing, text chunking, embedding generation, and vector indexing, improving retrieval relevance by 40% and contextual response accuracy by 35%.
- Built scalable backend services in **Python Flask** for document ingestion, vector indexing, and LLM orchestration, exposing functionality through a lightweight frontend interface and reducing query response latency by 30%.
- Engineered a modular **Selenium-Cucumber automation framework** using Page Object Model (POM) and behavior-driven testing, increasing regression coverage by 30%, while integrating **CI/CD pipelines** in **Jenkins** with **Maven**, **Gradle**, and **Git**-based version.
- Collaborated within an **Agile environment** to evaluate software systems for the Generative AI business unit using emerging technologies, AI-driven testing approaches, and quality metrics to strengthen continuous validation.

Software Development Engineer Associate Intern, HashedIn by Deloitte, Bengaluru, IN

March 2023 - June 2023

- Engineered a full-stack project management application using **Next.js**, **NestJS**, **React**, **Prisma ORM**, and **PostgreSQL**, supporting task management, issue tracking, story point assignment, and role-based collaboration across 50+ concurrent project items.
- Developed and maintained **RESTful APIs** using NestJS controller-service architecture and DTO-based validation, implementing **CRUD operations** for projects, tasks, and user management, improving API performance by 50% and code maintainability by 25%.
- Built responsive frontend components using React Hooks and Next.js routing while optimizing application state management, reducing client-side page reloads and improving navigation responsiveness by 35%.
- Designed relational data models, schema migrations, and optimized **PostgreSQL** queries using Prisma ORM, while implementing **JWT**, **OAuth 2.0**, and **RBAC** security mechanisms to improve scalability, data consistency, and multi-user collaboration.
- Validated API functionality using **Postman** and **Swagger**, ensuring reliable schemas and service quality throughout development cycles.

PROJECTS

Dynamic Traffic Density Analysis & Green Corridor Clearance – (KSCST & IISc Sponsored Research Project; Published Research Paper)

- Aimed to benefit traffic management by optimizing traffic signal timings with a specialized feature to detect emergency vehicles and priorities its passage by granting a green signal corridor.
- Implemented utilizing concepts of **TensorFlow**, **OpenCV**, **CNN**, **YOLO** & **Round-robin algorithms**. Real-time traffic inputs are broken down into frames and converted to grayscale to extract vehicle count and signal time durations are incremented/decremented accordingly.

Adaptive Virtual Assistant – (Industry-Sponsored Capstone Project for MyEdMaster, Herndon, VA)

- Engineered an adaptive AI-powered chatbot system using **Flask** and **Gemini API**, enabling personalized, context-aware responses and improving user engagement and response relevance.
- Designed and integrated a **semantic image retrieval system** processing 100K+ Wikimedia image using **OpenCLIP** and **ChromaDB**, enabling multimodal responses and improving contextual answer accuracy by 35%.
- Developed a high-throughput **asynchronous ingestion pipeline** using SQL dump processing, recursive CTE traversal, and multi-threaded processing (aiohttp + thread offloading), eliminating API rate limits and improving data processing efficiency by 50%.